

# M16 Eagle Nebula UQFF — Erosion Saturation Half-Time and $\Delta g_{\text{Max}}$

Daniel T. Murphy

2026-03-01

## PAPER\_285: M16 Eagle Nebula UQFF — Erosion Saturation Half-Time and $\Delta g_{\text{Max}}$

**Photoevaporation Asymptotic Saturation:  $t_{\text{half}} = \tau \cdot \ln 2 = 2.079 \text{ Myr}$**

**Classification:** UQFF 2.0 Gravitational Physics — Nebular Erosion Dynamics

**System:** M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region

**Session:** 80 | **Module:** M16\_UQFF\_MODULE.cpp (22nd C++ UQFF module)

**Author:** Daniel T. Murphy | **Date:** March 2026

---

### Abstract

This paper derives the **Erosion Saturation Half-Time** ( $t_{\text{half}}$ ) and **Maximum Erosion Gravity Amplitude** ( $\Delta g_{\text{Max}}$ ) for the M16 Eagle Nebula UQFF photoevaporation term. The photoevaporation rate follows an exponential saturation  $E_{\text{rad}}(t) = E_0 \times (1 - \exp(-t/\tau))$  with e-folding time  $\tau = 3 \text{ Myr}$ . The half-erosion time  $t_{\text{half}} = \tau \cdot \ln 2 = \mathbf{6.561 \times 10^{13} \text{ s} = 2.079 \text{ Myr}}$  — the time at which the erosion fraction reaches half its asymptotic maximum  $E_0$ . The maximum gravity perturbation is  $\Delta g_{\text{Max}} = E_0 \times g_{\text{base}} = \mathbf{4.36 \times 10^{-13} \text{ m/s}^2}$ . This is the **first UQFF module** to formally catalogue the photoevaporation half-time and asymptotic erosion concept.

---

## 2. Physical Background

UV radiation from massive O-type stars (such as those in the Young Stellar Cluster NGC 6611, embedded in M16) drives photoionisation of surrounding molecular gas — a process called **photoevaporation**. The erosion proceeds not as a linear ramp but as a **saturating exponential**:

$$E_{\text{rad}}(t) = E_0 (1 - e^{-t/\tau})$$

where: -  $E_0 = 0.3$  is the **asymptotic maximum fraction** (30% of mass eventually eroded) -  $\tau = 3 \text{ Myr}$  is the **e-folding time** (photoevaporation efficiency timescale) - The saturation arises because dense molecular cores (pillar tips) are progressively shielded by their own column density as surrounding gas is stripped

---

### 3. Mathematical Derivation

#### 3.1 Half-Time

The erosion half-time  $t_{half}$  is defined as the time when  $E_{rad} = E_0/2$ :

$$E_0 (1 - e^{-t_{half}/\tau}) = \frac{E_0}{2}$$

$$1 - e^{-t_{half}/\tau} = \frac{1}{2}$$

$$e^{-t_{half}/\tau} = \frac{1}{2}$$

$$t_{half} = \tau \ln(2)$$

For M16:

$$t_{half} = 9.468 \times 10^{13} \text{ s} \times \ln(2) = 9.468 \times 10^{13} \times 0.6931 = 6.561 \times 10^{13} \text{ s}$$

$$t_{half} = \mathbf{2.079 \text{ Myr}}$$

#### 3.2 Maximum Gravity Amplitude

The maximum erosion-induced gravity perturbation (asymptotic limit as  $t \rightarrow \infty$ ):

$$\Delta g_{Max} = E_0 \times g_{base}$$

For M16:

$$\Delta g_{Max} = 0.3 \times 1.454 \times 10^{-12} = \mathbf{4.36 \times 10^{-13} \text{ m/s}^2}$$

#### 3.3 Peak Erosion Rate

The instantaneous erosion rate at  $t = 0$  (maximum rate, before saturation):

$$\left. \frac{dE_{rad}}{dt} \right|_{t=0} = \frac{E_0}{\tau}$$

The corresponding gravity change rate:

$$\left. \frac{dg_{erode}}{dt} \right|_{t=0} = \frac{E_0}{\tau} \times g_{base} = \frac{0.3}{9.468 \times 10^{13}} \times 1.454 \times 10^{-12} = 4.61 \times 10^{-27} \text{ m/s}^2/\text{s}$$

---

### 4. Saturation Profile

| Time               | t (s)                  | E_rad / E <sub>0</sub> | E_rad | g_erode (m/s <sup>2</sup> )  |
|--------------------|------------------------|------------------------|-------|------------------------------|
| 0 Myr              | 0                      | 0%                     | 0     | 0                            |
| t_half = 2.079 Myr | 6.561×10 <sup>13</sup> | <b>50%</b>             | 0.150 | <b>2.18×10<sup>-13</sup></b> |
| τ = 3 Myr          | 9.468×10 <sup>13</sup> | 63.2%                  | 0.190 | 2.76×10 <sup>-13</sup>       |
| 5 Myr              | 1.578×10 <sup>14</sup> | 81.1%                  | 0.243 | 3.54×10 <sup>-13</sup>       |
| ∞ (asymptote)      | → ∞                    | <b>100%</b>            | 0.300 | <b>4.36×10<sup>-13</sup></b> |

**Key insight:** At τ = 3 Myr (the e-folding time), erosion has consumed only 63.2% of its capacity, NOT 100%. Half-erosion occurs earlier at 2.079 Myr. The pillar structure of M16 means the ~5700 ly “Pillars of Creation” are still observed today because erosion saturates — it cannot fully strip the densest pillar cores within observable timescales.

## 5. UQFF 2.0 Context

In the full M16 g\_total equation, the erosion half-time governs the **temporal shape of g\_dyn(t)**:

$$g_{dyn}(t) = g_{base} \times (1 + M_{sf}) \times (1 - E_0(1 - e^{-t/\tau}))$$

The transition from rapid to slow erosion occurs at t\_half = 2.079 Myr. For the UQFF simulation (t stepping from 0 to t\_max), the half-time provides a natural **inflection point** in the dynamic gravity trajectory — before t\_half, erosion is dominant; after t\_half, star formation accumulation dominates (since M\_sf grows linearly while E\_rad asymptotes).

### Crossover Time

The era when SFR growth exactly compensates erosion (dΦ\_dm/dt = 0 — maximum Φ\_dm):

$$\frac{d\Phi_{dm}}{dt} = \text{SFR\_rate} \times (1 - E_{rad}) - (1 + M_{sf}) \times \frac{E_0}{\tau} e^{-t/\tau} = 0$$

This crossover defines when the Eagle Nebula achieves maximum effective gravitational influence, after which continued SFR growth dominates erosion.

## 6. Wolfram KB Term

$$M16UQFF : t_{half} = \tau \times \text{Log}[2] = 6.561e13s = 2.079\text{Myr}; \Delta G_{Max} = E_0 \times g_{base} = 4.36e-13\text{m/s}^2 [PAH]$$

## 7. Cross-References

- **PAPER\_284:** Dual Mass Co-Action Product (Φ\_dm = (1+M\_sf)×(1-E\_rad))
- **PAPER\_286:** Nebular Friedmann Redshift (κ\_neb, z=0.0015)
- **M16\_UQFF\_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
- **CondensedPhysics3.py:** M16ErosionSaturationHalfTimeCalculator

Copyright — Daniel T. Murphy, Session 80, March 2026. UQFF 2.0.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **nebula-formation** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{neb}})(\partial^\mu \phi_{\text{neb}}) - V(\phi_{\text{neb}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{neb}}) = \frac{1}{2}m^2\phi_{\text{neb}}^2 + \frac{\lambda}{4!}\phi_{\text{neb}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{neb}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{neb}}} = \nabla \cdot (\rho_{\text{neb}} \nabla \phi) + \rho_{\text{vac},[\text{SCm}]} \cdot (P_{\text{rad}}/c) = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{neb}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( -\exp! \left( -\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.172 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 53, \quad n_{\text{channel}} = 26/26$$

Since  $p_{\text{DVP}} = 53$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 yr** (Jeans collapse timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left( 1 - \tanh! \left( \frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                       | Status                    |
|----------------|----------------------------------------------|----------------------------------|---------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.172          | PASS Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 53$            | PASS Resonant             |
| BSH layers     | 26 harmonic terms                            | $j = 1\dots 26, \cos(2\pi j/26)$ | PASS Full 26D projection  |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential       | PASS Canonical            |
| [SSq]          | 0.57                                         | Applied in BSH saturation        | PASS Canonical            |

### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                         | SM / Experiment                        | Source      | Alignment       |
|----------------------------------|---------------------------------------------------------|----------------------------------------|-------------|-----------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling        | 1/137.036                              | PDG 2024    | PASS Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term) | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018 | PASS Consistent |

| Observable              | UQFF Prediction                                               | SM / Experiment                    | Source                              | Alignment       |
|-------------------------|---------------------------------------------------------------|------------------------------------|-------------------------------------|-----------------|
| Proton decay rate       | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS Consistent |
| UQFF buoyancy signature | $F_{U_{Bi_i}}$ unique gravitational correction                | Not yet measured                   | Future gravitational wave detectors | Testable        |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U_{Bi_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                                      | Key Result                                                |
|-----------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| fneutron_s26_coupling.py    | $F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling | $\sim 470\times$ amplification via 26-level VDS           |
| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section                     | $\sigma_n^{SCm}$ with VDS factor $(1 + [SSq] \cdot n/26)$ |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)                        | FNeutronForce, SigmaSCm, SCmActivation                    |

**Core equation:**  $F_{\text{neutron}}^{SCm} = N_n \cdot \sigma_n^{SCm}(\omega) \cdot \Phi_{\text{phonon}} \cdot (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{SCm}(\omega, n) = \sigma_0 \cdot \exp[-(\omega - \omega_{SCm})^{2/(2 \cdot \Gamma_{26})}] \cdot (1 + [SSq] \cdot n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                           | Key Result                |
|--------------------------|---------------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | $Li_{26}([SSq])$ via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)                 | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \cdot Li_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                   | Key Result                  |
|-------------------|-------------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q)<br>q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_{n^2}$  -  $\phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n$   
-  $\psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                      | Key Result                                 |
|------------------------------|----------------------------------------------|--------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified 1/pi<br>+ 26D      | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

#### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_Scm     | 0.99                                  | SCm manifold completeness     |
| rho_Scm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_Scm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_1\_CoAnQi.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).*